# Harsh Bhatter

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## EDUCATION

**Jadavpur University,** Kolkata, India — *October 2023 - June 2027*
B.E in Power Engineering (E.E)
CGPA: 8.1 (upto 5th semester)

**Purwanchal Vidyamandir,** Kolkata, India — *2022*
ISC Percentage: 93.75%

**Purwanchal Vidyamandir,** Kolkata, India — *2020*
ICSE Percentage: 92.2%

## EXPERIENCE

**Software Development Intern, RawRecruit** — *July 2026 – August 2026*

- Developed and maintained RESTful APIs using Node.js, Express.js, and MongoDB for production-facing features.
- Built and enhanced a data normalization pipeline for company and college records using alias matching, Fuse.js fuzzy search, confidence scoring, and canonical ID mapping.
- Conceived and built the backend for a FOMO-driven engagement feature — surfacing trending job postings, alumni who posted a job, and alumni who received a referral — to drive re-engagement on the platform.
- Worked with JWT-based authentication, including access and refresh token workflows, for secure session management.
- Modified MongoDB schemas and backend services to support merge operations and canonical data management.
- Diagnosed and resolved production bugs across backend APIs, database logic, authentication, and frontend integration.
- Collaborated using Git, pull requests, and code reviews in a production environment.

## PROJECTS

**E-Commerce Web App** | React, Spring Boot, MySQL, OAuth2, JWT, Razorpay | Link

- Built a full-stack e-commerce platform featuring product catalogue, cart management, order summary, order management, Google OAuth2 login, JWT authentication, and Razorpay payment integration.
- Implemented a concurrency-safe inventory reservation workflow using locking to prevent overselling during simultaneous checkouts; validated through concurrent load testing with Apache JMeter.
- Designed extensible coupon and notification modules, enabling new coupon types, discount rules, and notification channels to be added with minimal code changes; implemented email notifications for order events via JavaMail.
- Reduced product catalogue API average latency from 3.4s to 51ms by replacing heavy entity responses with lightweight DTOs and dedicated image endpoints; validated under concurrent load using Apache JMeter.
- Deployed backend on Render and frontend on Netlify with MySQL hosted on Aiven, resolving cross-service configuration issues (CORS, HTTPS, timezone) across distributed deployments.

**API Rate Limiter** | Java, Spring Boot

- Built a configurable API rate-limiting system using the Strategy Design Pattern, supporting Token Bucket, Fixed Window, and Sliding Window algorithms for interchangeable traffic-control policies.
- Integrated rate limiting at the servlet filter layer via Spring Boot's OncePerRequestFilter, intercepting requests before they reach controllers — with per-client throttling using thread-safe concurrent data structures.
- Implemented HTTP 429 responses with rate-limit and retry headers, and endpoint-specific configurable limits so different APIs can use different strategies and thresholds.

**Whiteboard** | HTML, CSS, Vanilla JavaScript | Link

- Designed, built, and tested a fully functional whiteboard application with pencil, eraser, sticky notes, image insertion, and board download features.
- Implemented undo-redo feature using arrays as a stack by storing positions.
- Gained hands-on experience in DOM manipulation, canvas API, and frontend design principles.

## TECHNICAL SKILLS

**Languages:** Java, SQL
**Frontend:** React, HTML, CSS
**Backend:** Spring Boot, Spring MVC, Hibernate, JPA, RESTful APIs
**Databases:** MySQL, MongoDB
**Tools:** Git, GitHub, Postman, Apache JMeter
**Coursework:** Data Structures and Algorithms, Object Oriented Programming, Operating System, Computer Networks

## ACHIEVEMENTS

- Secured Global Rank 80 among 33,660 participants in CodeChef Starters 202.
- Earned Knight badge on LeetCode (Max – 1980). [profile]
- Achieved a rating of 1332 (Pupil) on Codeforces. [profile]
- Achieved a 3-star rating on CodeChef. [profile]
- Finalist in Competitive Programming events at Srijan 2024 & 2025, the annual tech fest of Jadavpur University.
- Secured General Rank 1595 in WBJEE.